

RESEARCH PROPOSAL: ADVANCING ARTIFICIAL INTELLIGENCE IN ACADEMIC ENVIRONMENTS

The rapid evolution of Artificial Intelligence and Computer Vision is transforming the way modern educational institutions operate. This document outlines a comprehensive framework for integrating advanced neural networks into university laboratories and student assessment systems. By utilizing sophisticated computer vision algorithms, we can create smarter, interactive learning environments that visually adapt to individual student needs and provide real-time feedback during practical lab assignments. The proposed technical system relies on robust Python-based backend architectures, specifically utilizing FastAPI for fast data handling and seamless integration with various front-end interfaces. While the implementation of such automated systems brings immense benefits to the education and technology sectors, it is crucial to maintain strict privacy standards to ensure that sensitive academic data remains secure from any potential leak. Fortunately, we do not anticipate any critical infrastructure issues during the initial deployment phase. The primary goal of this initiative is to empower senior computer engineering students to develop innovative, AI-driven graduation thesis projects. Providing them with hands-on experience in these fields will ultimately prepare them for voluntary internships and professional careers in leading tech companies. Furthermore, the research will explore how specialized freelance business models—such as automated digital event platforms—can be supported through these same foundational AI technologies.